

Research Paper

Research on thermal management of 3D-ICs assisted by deep learning

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ABSTRACT

Compared with integrated circuits based on through silicon via (TSV), monolithic inter-tier via (MIV) has been identified as a critical technique to enable three dimensional (3D) integration due to its ultra-small size and relatively superior electrical performance, which allows ultra-high integration density. However, the interconnection of monolithic 3D (M3D) design is more prone to electromigration and stress migration. Severe crosstalk during signal transmission and thermal stress at high temperatures have serious limitations on system performance. In this paper, we focus on the COMSOL Multi-physics software, which can solve multi-field problems, to study the crosstalk problem and its thermal stress problem in MIV structures and analyze the crosstalk effects and temperature stress changes of MIV under different physical coupling conditions. An MIV array based on electrical-thermal-mechanical multi-field coupling was proposed, and the temperature and stress were analyzed by finite element analysis software. Additionally, an artificial neural network scheme is proposed that uses MATLAB to train temperature and stress data to predict the stress values of MIV. Experimental results show that the proposed prediction model using a genetic algorithm to optimize the BP Neural Network (GABP) has a 23.3 % higher prediction accuracy than that of a general BP neural network.

1. Introduction

The integrated circuit (IC) industry has seen remarkable progress since the 1970s, effectively advancing Moore's law [1]. However, progress in this field has hindered in recent times due to the restrictions posed by the feature size of transistors. In order to continue advancing, 3D IC technology has emerged as a promising solution for increasing integration density [2]. TSVs have become a key component of 3D integration, facilitating interconnections and reducing chip area while enabling heterogeneous integration. TSVs realize the interconnection of multi-layer stacked chips through the substrate, minimize chip area, increase data transmission bandwidth, and are an important means to achieve 3D integration [3]. However, TSVs in the above research still present challenges to 3D ICs due to their large size.

The M3D technology is distinct from the TSV-based circuits used for wafer stacking [4]. M3D employs fine-grained MIVs to achieve inter-layer interconnections [5]. Its structure closely resembles that of conventional metal vias, as shown in Fig. 1. The basic process flow of MIV includes chip cutting and surface treatment; pattern design and lithography; etching; metal filling; and handling and testing. This process

involves sequentially stacking M3D integrations on top of each other and entails the cutting of layers built on a single semiconductor wafer into a three-dimensional integrated circuit [6,7]. Due to the nanoscale structure size of MIV, the chip can accommodate a large number of vertical MIV connections, significantly enhancing the integration level compared to microscale TSV-connected 3D ICs [8]. However, from a hardware security perspective, this technology also presents new challenges. M3Ds, which experience severe heat dissipation, are more susceptible to attacks caused by thermomechanical deformation. With the extensive development in the industrial field, MIV as the medium for signal transmission, encounters serious heat dissipation issues. The differing thermal conductivity of materials hinders effective heat dispersion, exacerbating the thermal stress problem [9]. As such, the continued advancement of M3D integration faces various obstacles. Consequently, the industry urgently requires more efficient heat dissipation designs and solutions to enhance device performance.

In recent years, the ongoing advancement of artificial intelligence technology has played a pivotal role in circuit design and optimization. By employing machine learning and deep learning algorithms, the structure of circuits can be simulated, analyzed, and optimized based on

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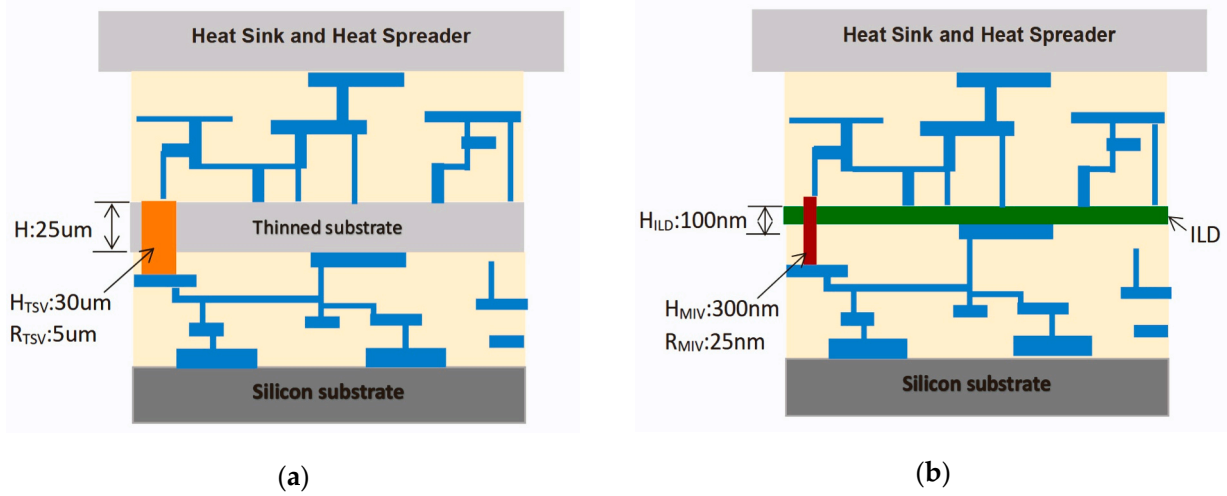


Fig. 1. Two-layer 3D-IC structure: (a) TSV 3D-IC; (b) MIV 3D-IC.

specific performance indicators, which significantly enhances the system's performance reliability. Numerous international studies have explored the integration of 3D ICs with artificial intelligence [10]. Ning et al. combined deep learning based on convolutional neural networks (CNNs) with three-dimensional X-ray tomography, obtaining accurate scanning images by training the artificial intelligence machine with a reliable image acquisition database [11]. Additionally, Y et al. utilized unsupervised graph learning with graph neural networks (GNNs) and advanced machine learning techniques for floorplanning, resulting in substantial improvements in performance, wirelength, and per-cycle energy [12]. Furthermore, Zhang et al. developed a novel rapid stress analysis method to increase the runtime reliability of 3D ICs. Their approach utilized an artificial neural network (ANN) model trained to pinpoint the maximum stress for each TSV, integrating this with dynamic thermal management (DTM) techniques to enhance the reliability of 3D ICs [13].

In this paper, we investigate the impact of single-factor variations in MIV on crosstalk response under the influence of multiple physical fields, as well as the changes in temperature and stress responses. Due to the significant disparity in the coefficient of thermal expansion (CTE) between copper and silicon materials, high temperatures can induce stress alterations around the MIV, potentially affecting equipment performance or precipitating crack propagation, thereby reducing the lifespan of the 3D IC. We analyze the MIV's temperature and stress fluctuations in 3D ICs and introduce a thermal stress analysis model for MIV that is predicated upon electro-thermo-mechanical multi-field coupling. To capture stress data, diverse temperatures are applied to the model. The surrounding temperature distribution data for each MIV serves as the model's input, with critical stress metrics, including maximum stress around each MIV, as the output. This model is then refined through training with a backpropagation neural network optimized by a genetic algorithm. The experimental results demonstrate that, relative to a standard BP neural network, the enhanced stress model incorporating the genetic algorithm exhibits superior accuracy in stress estimation and reduced computational error.

2. Methods, results and discussions

In this section, we model three-dimensional integrated circuits and analyze the effects of changes in individual factors on electromagnetic crosstalk. MIV plays the role of transmitting signals in 3D integration. But with the continuous improvement of the requirements for integrated systems, its work frequency is also gradually increasing. In the 3D ICs, when transmitting signals to each other, hundreds of MIVs crosstalk is getting stronger and stronger, resulting in the crosstalk phenomenon,

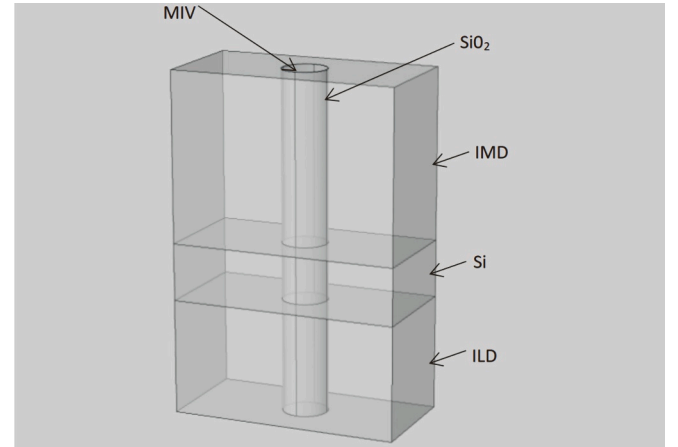


Fig. 2. Model diagram based on MIV.

which affects the transmission quality and reduces the integrity of the signal [14]. Therefore, in order to improve the electrical reliability of MIVs, discussion of the influencing factors of crosstalk is urgent and necessary.

2.1. Model of M3D ICs

Fig. 2 illustrates a 3D electrical model of a MIV that we developed using the multiphysics simulation software COMSOL. As depicted, the conductive filler of the MIV consists of copper (Cu), encased within a SiO₂ insulating layer. The MIV structure extends through both the thin silicon layer known as the inter-metal dielectric (IMD) and the inter-layer dielectric (ILD). The specific process parameters for the MIV construction are detailed in Table 1.

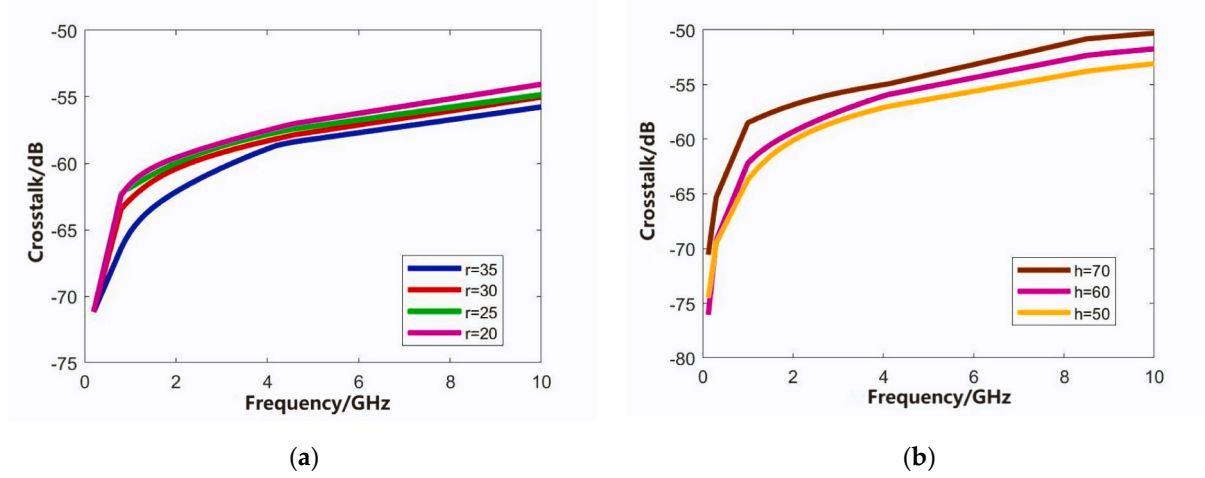
2.2. Electromagnetic physical field crosstalk analysis

In the simulation of electromagnetic crosstalk, the problem is addressed by generating a mesh, defining boundary conditions, and incorporating an electromagnetic field. A specific temperature is set at the model's base, and a predefined heat dissipation coefficient is applied at the top. This study analyzes the effects that alterations in the radius, length, height, and filling material of the MIV have on the system's crosstalk. Simulations are conducted to determine the magnitude of crosstalk associated with these structural parameters.

Table 1

Thermal simulation parameter setting.

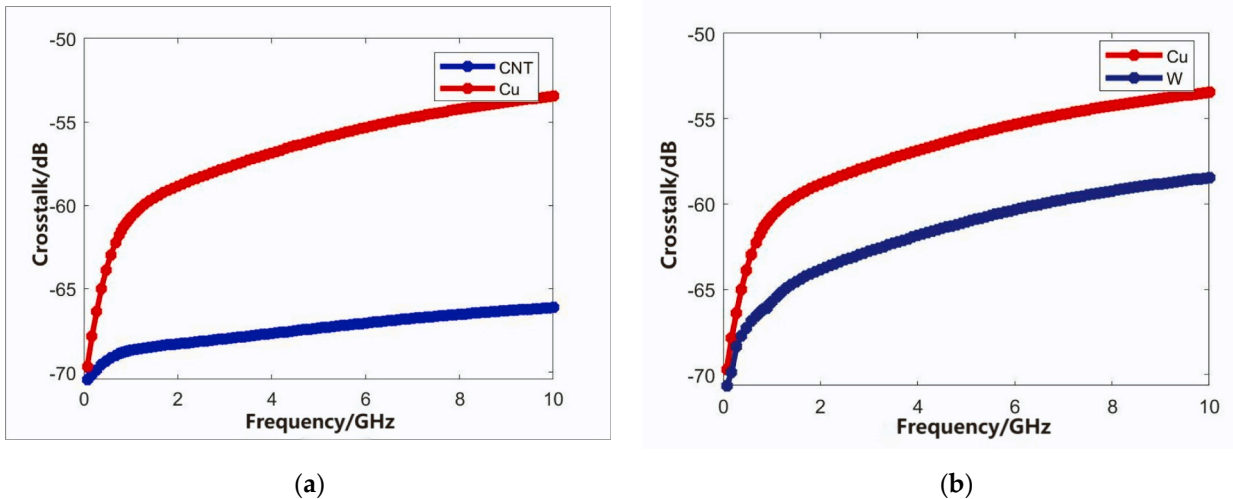
Parameter	Materials	Density (kg/m ³)	Young's modulus (Pa)	Heat conductivity coefficient (W/m·K)	Coefficient of thermal expansion (K ⁻¹)	Electrical conductivity (S/m)
MIV	Cu/CNT	8960/1400	$1 \times 10^{11}/1 \times 10^{12}$	401/3000	$17 \times 10^{-6}/4.15 \times 10^{-6}$	$5.71 \times 10^7/10^6$
MIV	W	1.93×10^4	3.66×10^5	163	4.4×10^{-6}	1.76×10^7
Si	Silicon	2329	1.9×10^{11}	130	4.15×10^{-6}	—
ILD/IMD	SiO ₂	2200	7.3×10^{10}	1.4	0.5×10^{-6}	—

**Fig. 3.** The crosstalk varies with the MIV parameter: (a) crosstalk with radius; (b) crosstalk with ILD thickness.

MIV functions similarly to a wire when utilized for signal transmission [15,16]. Variations in its radius and height influence the cross-sectional area and length through which the charge is conducted, thereby impacting signal transmission. In the electromagnetic crosstalk simulation, we verify the influence of MIV with different radius of 20 nm, 25 nm, 30 nm and 35 nm on crosstalk. The influence of ILD thickness on crosstalk is verified by changing the thickness of ILD to 50 nm, 60 nm and 70 nm respectively. Since MIV penetrates the ILD layer, the height of MIV also changes accordingly. The specific changes are shown in the Fig. 3. The role of the ILD insulation layer is to isolate the silicon plate from the filled conductive material to prevent current leakage. The high dielectric constant of silica shields the signal interference between layers, thereby reducing the crosstalk between signals and ensuring the reliability of the circuit [17].

From Fig. 3(a), it is observable that with all else being constant, crosstalk intensifies as frequency escalates. Examining the impact of varying MIV radius, it is evident that a larger radius correlates with reduced crosstalk. This reduction is attributed to the larger diameter diminishing the coupling effect between a transmission line and its adjacent lines, which in turn mitigates crosstalk. In Fig. 3(b), when comparing the MIV structures with different filler sizes against ILD thickness, there's a clear trend indicating that as frequency increases, so does crosstalk. However, a thinner ILD corresponds to lower crosstalk and improved transmission characteristics. Conversely, a thicker ILD augments the separation between device wiring, enhances coupling between transmission lines, and thus proliferates crosstalk.

The selection of fill materials for MIV is a critical factor in the performance of three-dimensional integrated systems. Cu is extensively

**Fig. 4.** The crosstalk caused by filling different materials in MIV. (a) Crosstalk comparison of MIV filled CNT and Cu respectively, (b) crosstalk comparison of MIV filled W and Cu respectively.

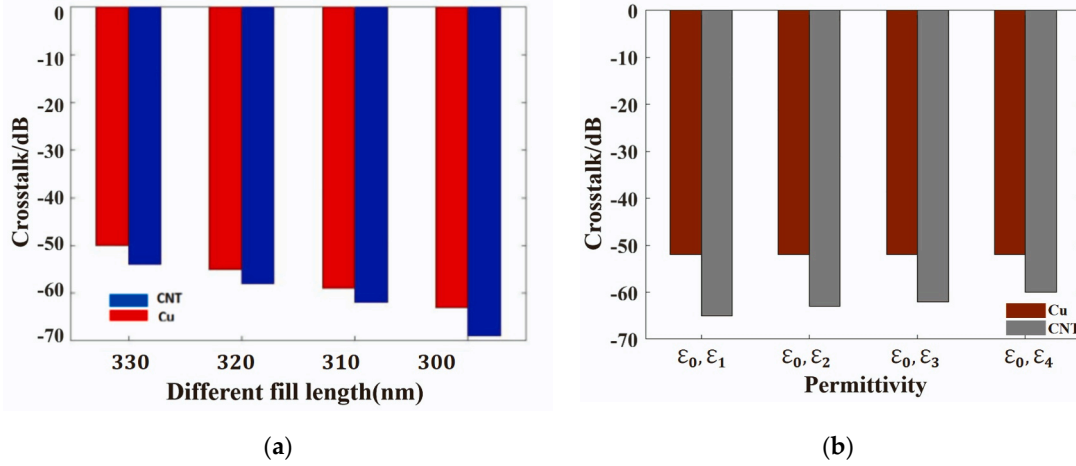


Fig. 5. Crosstalk varies: (a) the crosstalk caused by different fill length, (b) the crosstalk caused by different permittivity.

used as a fill material in MIVs due to its low resistivity and high electrical conductivity. Carbon nanotubes (CNT) are considered promising interconnect materials on account of their distinctive electrical and thermal characteristics, including their high elasticity and toughness, which contribute to lower resistivity and improved conductivity. The electrical properties of CNT vary based on their diameter, wall thickness, and structural configuration, resulting in different conductivity levels among various CNT. For an optimal result, the MIV filling process with CNT and copper should be devoid of any loss, ensuring that the material adheres well to the substrate. A widely used technique involves the synthesis of Cu-filled MIV through a chemical vapor deposition or physical vapor deposition [18,19]. Growing CNT based on low-pressure chemical vapor deposition (LPCVD) [20] and solution-based deposition [21] pave the way towards the CNT as a filler material in MIV for M3D ICs [22].

In the subsequent simulations, the impact of various filler materials on crosstalk within MIVs will be examined. This analysis will be conducted by altering only the filler material inside the MIVs. Adjusting the fill heights of both Cu and CNT will grant insight into how these variations influence the extent of crosstalk.

Fig. 4(a) demonstrates a clear trend where crosstalk intensifies progressively as the frequency rises, and CNT filled MIVs exhibit less crosstalk compared to Cu filled MIVs. The superior electrical characteristics of CNTs, coupled with their low resistivity and low dielectric constant, contribute to their effectiveness in mitigating some of the losses typical in MIVs. Consequently, in the context of high-frequency applications, CNT MIVs exhibit a greater capability for efficient energy transfer and enhanced interface matching. Fig. 4(b) illustrates that tungsten (W) fillers result in lower crosstalk compared to copper fillers. Despite copper's higher electrical conductivity, which typically suggests

superior conductivity for copper fillings compared to tungsten, the transmission of high-frequency signals is adversely affected by copper's resistance and inductance properties. These characteristics can lead to signal reflection and attenuation, thereby contributing to more significant levels of crosstalk.

Fig. 5(a) indicates that as the height of the filler material within the MIV decreases, so does the crosstalk. This is because as the fill height increases, so does the area of electromagnetic coupling between adjacent lines. This electromagnetic interaction interferes with adjacent lines through the interaction of electric and magnetic fields. In summary, across the scenarios presented, CNT tends to cause less crosstalk than Cu, offering a more advantageous solution for applications where mitigating crosstalk is essential. Furthermore, Fig. 5(b) illustrates the variation of crosstalk by changing only the permittivity of the material. By keeping the permittivity of $\text{Cu}\epsilon_0$ constant, the permittivity of CNT was varied from $\epsilon_1 = 4$, $\epsilon_2 = 6$, $\epsilon_3 = 8$, to $\epsilon_4 = 10$. It can be observed from the figure that the crosstalk increases gradually with the increase of CNT's permittivity, but the overall crosstalk is still lower than that of Cu.

The following Fig. 6 shows the temperature gradients above and below the MIV filled with copper and CNT, respectively. In general, the smaller the temperature gradient is, the better, because the smaller temperature gradient means that the temperature distribution in the whole chip is more uniform, which can reduce the thermal distortion and fault in the high temperature region. CNT have excellent thermal conductivity, which can quickly conduct heat and balance the temperature distribution. The low density and flexibility of the CNT material can reduce the weight and stress of MIV, reduce the thermal stress and reduce the temperature gradient.

In cases where the filler material is CNT, the change in separation

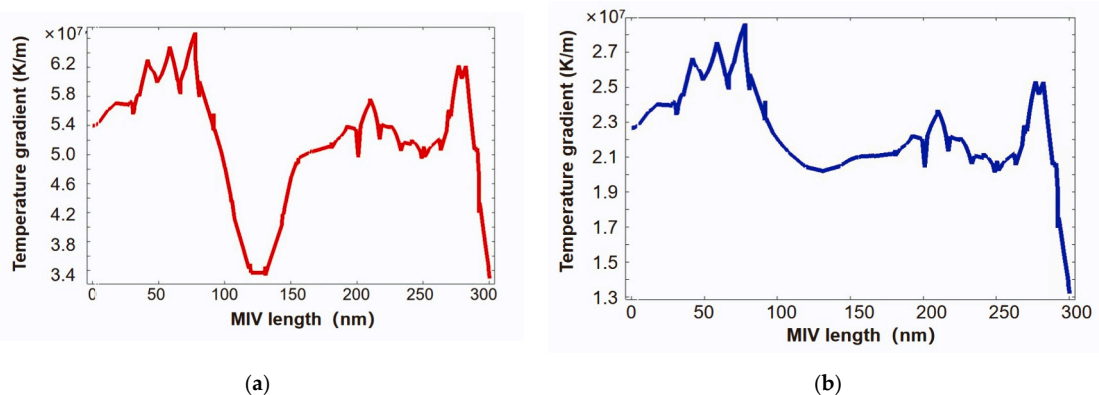


Fig. 6. The temperature gradients changes of different materials filled in MIV: (a) copper-filled MIV; (b) CNT-filled MIV.

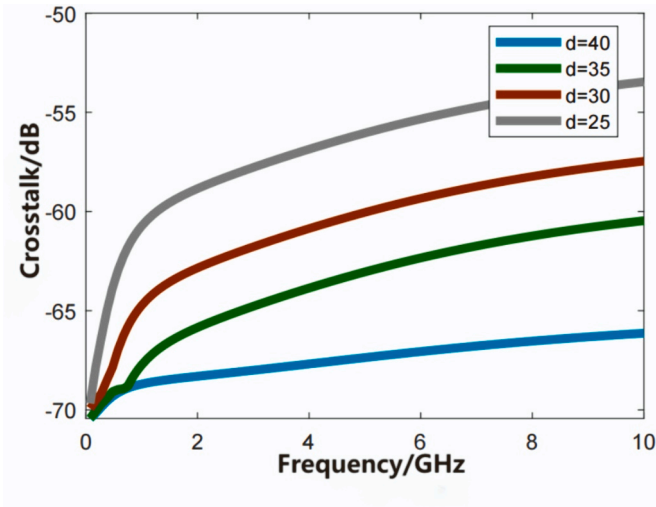


Fig. 7. Crosstalk variation at different spacing.

distance of the MIV significantly affects crosstalk; larger spacing results in a larger silicon substrate size between MIVs and a longer path for signal crosstalk between MIVs. This section examines the variation in crosstalk that occurs when MIVs are positioned adjacent to each other at separation distances of 25 nm, 30 nm, 35 nm, and 40 nm, respectively.

From Fig. 7 we can see that the crosstalk between MIVs decreases as the spacing between MIVs increases. As the spacing of the MIVs

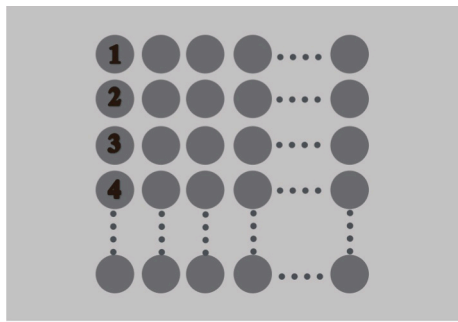
increases, which means the distance between the two MIVs is larger, the signal, being electromagnetic during transmission, the interaction between the fields weakens, which reduces the degree of crosstalk. This is because the larger spacing reduces the coupling strength of the electromagnetic field and reduces the possibility of mutual interference between signals.

The effect of MIV spacing on crosstalk is analyzed above. In practical applications, MIVs often appear in the form of arrays, and we construct MIV arrays and extract MIVs in the vertical direction of 1–4 to explore the crosstalk effect when they are coupled with each other.

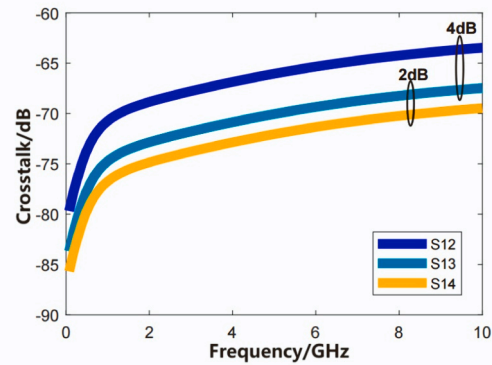
The specific results are shown in the figure below: from the Fig. 8, we can see that the interference of signals 3, 4, and 5 on signal 1 gradually decreases because the spacing decreases the crosstalk between the signals. Inserting other MIVs between two MIV also affects the crosstalk between them. For example, when analyzing S14, it is affected by the signals of codes 2 and 3 because the material that fills the MIV vias is mostly conductor metals such as Cu, which absorbs the electromagnetic signal in the silicon substrate and reduces crosstalk between MIVs, acting as a shield.

2.3. Electro-thermal-mechanical multi-physics coupling analysis

In addition to the problem of crosstalk, the thermal stress of MIVs in 3D integrated circuits is also worth discussing. In this section, we couple a variety of physics, including electric, thermal, and force fields, to analyze the various variations caused by different MIV filling materials, as well as the effects of changes in MIV size on temperature and stress generation. Solid stresses in Cartesian coordinates can be expressed as [23]:

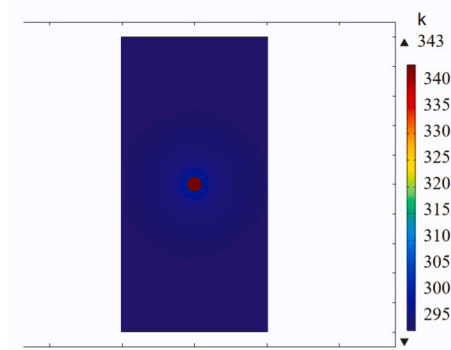


(a)

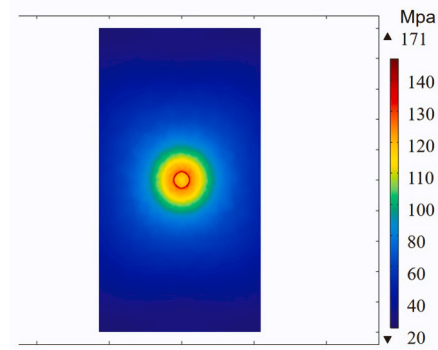


(b)

Fig. 8. (a) Schematic diagram of different locations of the MIV; (b) crosstalk at different locations of the MIV.



(a)



(b)

Fig. 9. The temperature and stress distribution of MIV: (a) temperature distribution of MIV cross section; (b) stress distribution of MIV cross section.

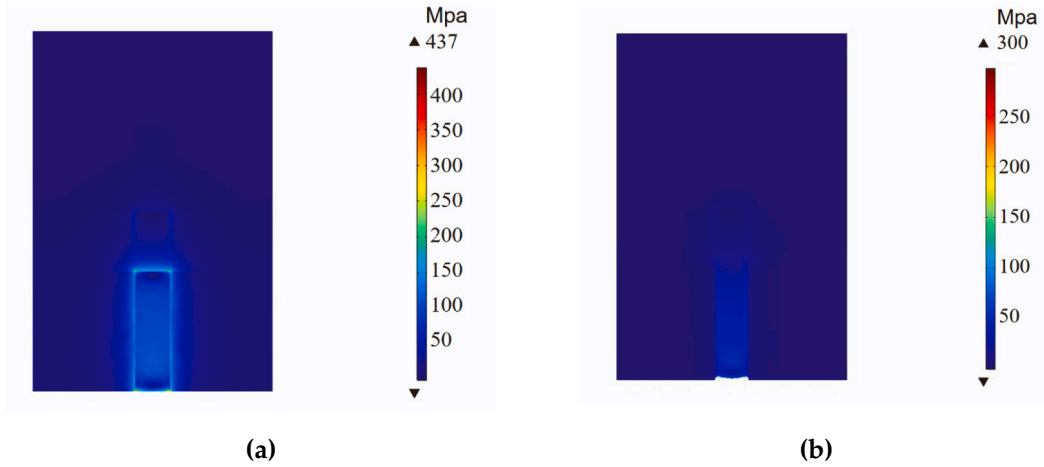


Fig. 10. The stress changes of different materials filled in MIV: (a) copper-filled MIV; (b) CNT-filled MIV.

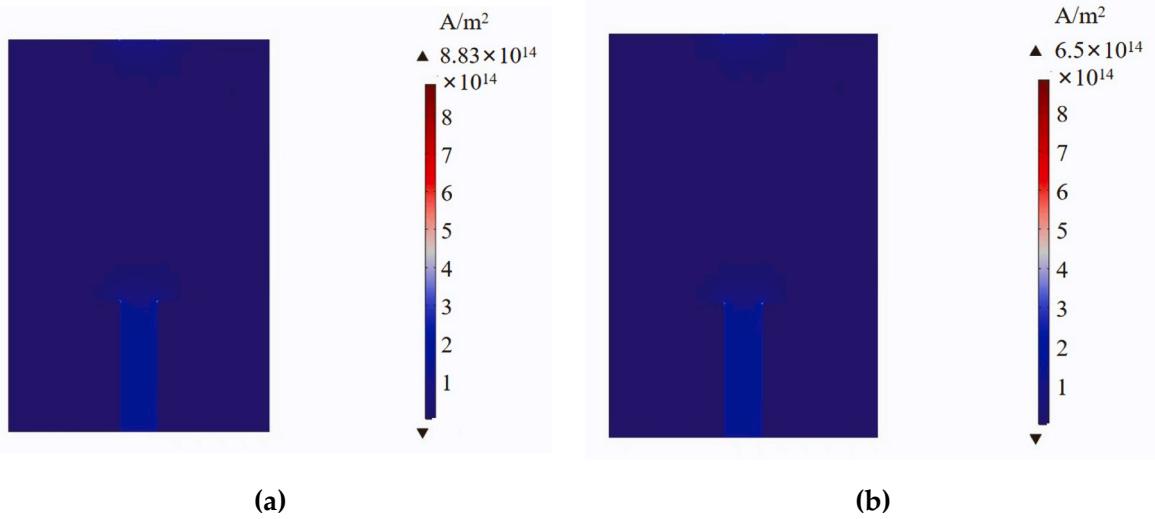


Fig. 11. The current density changes of different materials filled in MIV: (a) copper-filled MIV; (b) CNT-filled MIV.

$$\begin{cases} f_z = \frac{E\alpha}{1-2\gamma} \cdot \frac{\partial T}{\partial z} - k_z \\ f_y = \frac{E\alpha}{1-2\gamma} \cdot \frac{\partial T}{\partial y} - k_y \\ f_x = \frac{E\alpha}{1-2\gamma} \cdot \frac{\partial T}{\partial x} - k_x \end{cases} \quad (1)$$

In Eq. (1), the force in the x, y, and z directions are denoted as f_x , f_y , and f_z , respectively, while E represents the elastic modulus. Poisson's ratio is denoted by γ , and α represents the thermal expansion coefficient. Moreover, T denotes temperature in the given context. From the formula we can observe the f_x f_y f_z change with temperature T .

Then, we use the finite element simulation software COMSOL for simulation verification. The simulation model corresponds to the setup described earlier. The finite element model of the entire chip is placed in an environment reflecting room temperature, which is 293.15 K. The bottom surface of the model is established as a fixed constraint, and all surfaces except the bottom and top are defined as insulating boundaries.

We analyze the temperature and stress distribution of the 3D integrated chip with MIV structure as shown in Fig. 9. We can see that from the simulation results that the peak temperature of the MIV connected to the top is mainly composed of copper with high thermal conductivity. The maximum stress of MIV appears at the junction of MIV and

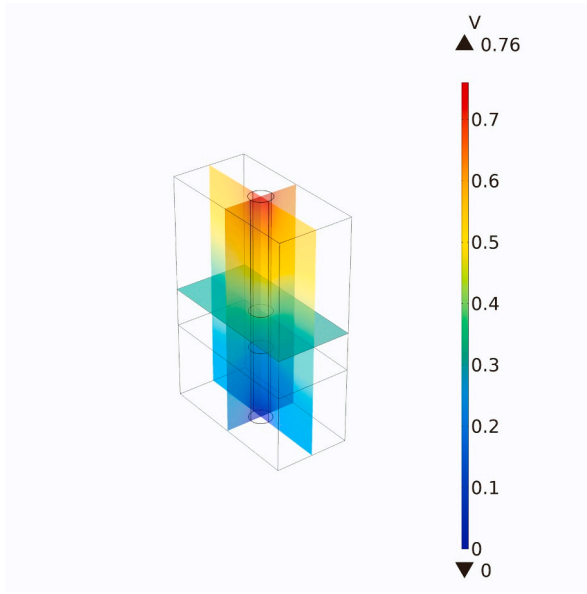
insulation layer. Then, when the position is far away from the MIV center, the stress gradually decreases, which is caused by the coefficient of thermal expansion (coefficient of thermal expansion, CTE) difference between the filling materials of MIV [24].

Using the circuit model described above, we have determined the stress and current density at various positions along the longitudinal interface of the MIV through-hole when the MIV filling materials are copper and CNT. The specific results are illustrated in Figs. 10 and 11.

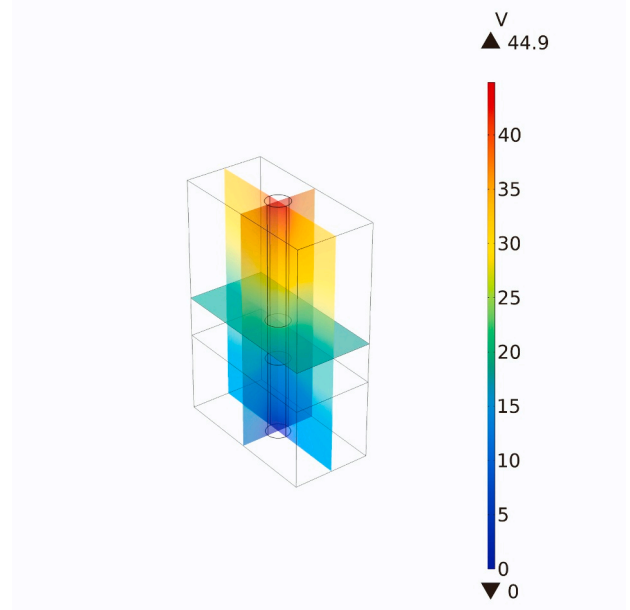
From Fig. 10, we observe that the equivalent stress values generated with copper filling and CNT filling are 437 MPa and 300 MPa, respectively. Since the CTE of CNTs is relatively lower than that of Cu, CNTs help dissipate heat from the channel more effectively, resulting in reduced stress during operation.

From the Fig. 11 we can see that the current density of MIV based on copper and CNT is 8.83×10^{14} A/m² and 6.5×10^{14} A/m², respectively. The current density in the MIV with CNT as the filling material is lower compared to that in the copper-based MIV. An increase in current density correlates with an increase in energy loss within the circuit. Additionally, current density influences heat dissipation, which can impact circuit or device performance. Consequently, the lower current density observed with CNT filling contributes to better circuit performance.

The inductor within the MIV structure generates Joule heat during the energization process, leading to an increase in temperature across the entire MIV inductor. Owing to the disparities in thermal expansion

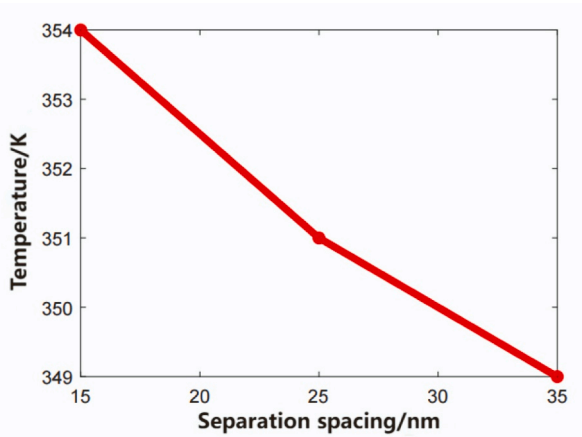


(a)

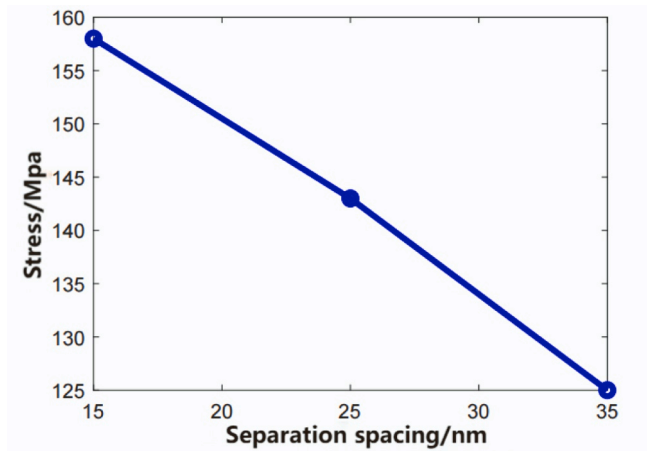


(b)

Fig. 12. The inductance potential changes of different materials filled in MIV: (a) copper-filled MIV; (b) CNT-filled MIV.



(a)



(b)

Fig. 13. Temperature and stress changes at different spacing: (a) temperature change; (b) stress change.

coefficients across different materials and structures, a variety of thermal stresses and deformations arise around the inductor, influencing the electrical characteristics of the entire structure. The distribution of electric potential within the MIV structure is depicted in Fig. 12.

The electrical potential in CNTs is higher than that in copper, which could prove advantageous for the MIV system. CNTs exhibit very high conductivity and electron mobility, meaning electrons can flow freely within them, creating regions with strong electric fields. This characteristic enables the formation of a relatively high electric field, contributing to a greater electrical potential in the MIV utilizing CNTs compared to copper. Furthermore, CNTs can also exhibit significant electron density changes at the nano-scale in scenarios of confined electron channels, which contributes to the higher electric potential of CNTs relative to copper.

In this paper, the temperature and stress changes of MIVs are

analyzed when the separation spacing between MIVs varies by 15 nm, 25 nm, and 35 nm, respectively, and the results are shown in Fig. 13.

The experimental results reveal that with increasing MIV spacing, both temperature and mechanical stress decrease. The vertical stacking of circuits poses heat dissipation challenges, leading to MIVs being subjected to high-temperature zones when situated too closely. Conversely, increased spacing around each MIV offers greater thermal resistance, thereby mitigating the temperature impact when the MIVs are more distant from one another. Close proximity of MIVs can exert stress on adjacent chips and transistors due to their operation and heat generation, potentially impairing performance. Such stress may induce deformation in transistors, resulting in diminished performance or failure.

In the M3D IC, the arrangement of thinner ILD and ultra-small spacing MIVs helps to achieve effective vertical heat flux. In addition,

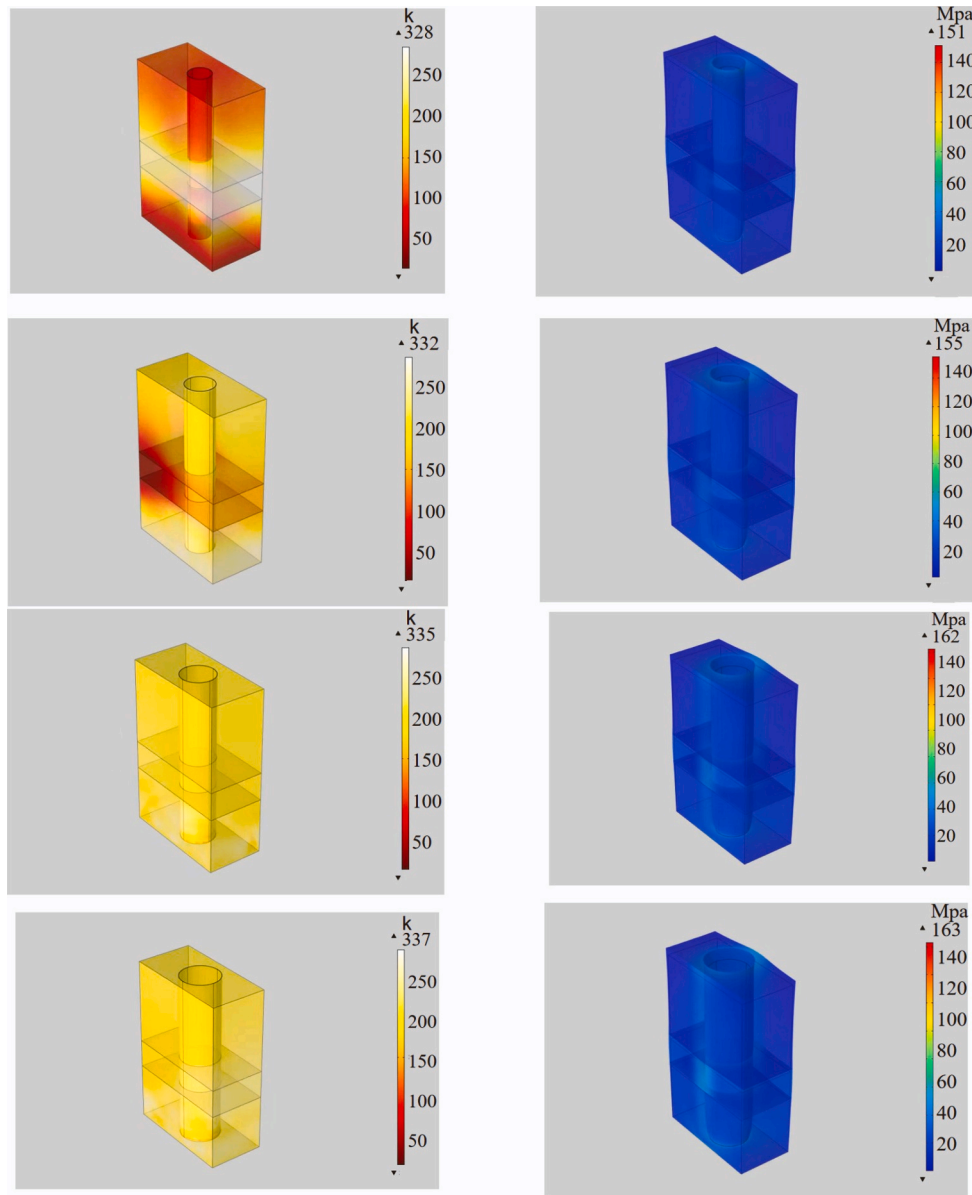


Fig. 14. Temperature and stress at different radius of MIV.

Table 2

The change of temperature and stress under different radius conditions.

	Radius/nm			
	15	20	25	30
Temperature/K	328	332	335	337
Stress/MPa	151	155	162	163

due to the very thin device layer, the transverse heat conduction is almost zero, which can be ignored. Therefore, the vertical heat conduction is stronger than the transverse heat conduction. The inherent characteristics of this strong vertical thermal coupling and finite lateral coupling reduce the maximum temperature of the M3D integrated circuit chip [25]. We verify the influence of different MIV radius of 15 nm, 20 nm, 25 nm and 30 nm, the results are shown in Fig. 14; the variations of temperature and stress with radius are shown in Table 2. Verify the effects of ILD height changes of 90 nm, 100 nm, 110 nm and 120 nm on the temperature and thermal stress of the whole 3D circuit.

Table 2 show that the temperature and stress increase with the

increase of MIV radius. When the radius is larger, the temperature will increase accordingly. This is because, in the process of heat transfer, the thermal resistance is proportional to the square of the radius. That is to say, as the radius increases, the heat in the microelectronic chip will be more difficult to disperse. In this case, more heat will be accumulated, resulting in an increase in the temperature inside the chip, indicating that the large-diameter MIV is not conducive to the heat dissipation of the system and is more prone to thermal reliability problems. Fig. 15 shows the influence of the thickness change of the ILD medium layer on the temperature and stress of the MIV.

From Fig. 15, we can see that the increase of ILD thickness will lead to the increase of temperature and stress. This is attributed that the increase in the thickness of the ILD leads to an increase in thermal resistance, making it more difficult for heat to propagate inside the chip. During the transmission process, heat will accumulate in the ILD, resulting in an increase in local temperature. Therefore, when the thickness of ILD increases, the temperature in MIV will increase accordingly.

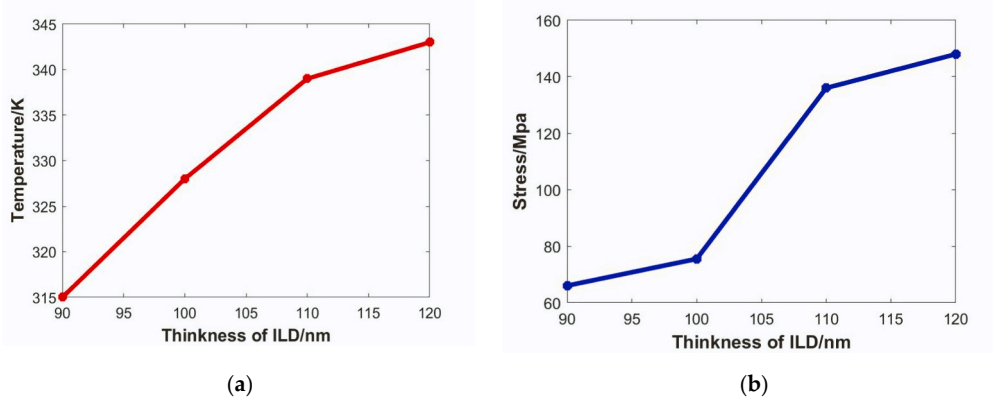


Fig. 15. The effect of different thickness of ILD change.

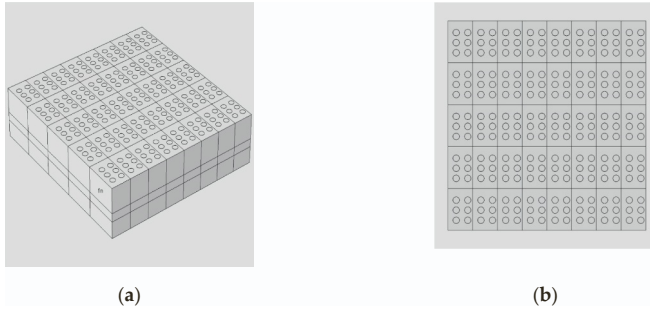


Fig. 16. COMSOL three-dimensional integrated circuit model: (a) side view; (b) top view.

3. Stress analysis of MIV structure based on artificial neural network

MIV technology is introduced into 3D ICs to enable chip stacking, aiming to enhance chip integration. This technology is crucial in addressing the challenges posed by the breakdown of Moore's Law. Silicon has a thermal conductivity of 130 W/m·K, while copper boasts a thermal conductivity of 401 W/m·K, which is more than three times that of silicon. This significant difference in thermal conductivity results in a large temperature gradient in the stress model. [26]. The CTE of silicon and Cu are $4.15 \times 10^{-6} \text{ K}^{-1}$ and $17 \times 10^{-6} \text{ K}^{-1}$, respectively. When the temperature has the same degree of change, the expansion of copper will experience larger stress compared to the silicon, resulting in chip fracture and interface delamination [27]. We carried out thermal stress analysis and modeling of 3D ICs and analyze the change of MIV stress in 3D IC caused by temperature in detail. We established the MIV stress prediction model in COMSOL and carried out the simulation. Finally, the artificial neural network is used to predict and analyze the data in

MATLAB, verifying the accuracy and reliability of the prediction model.

3.1. Establishment of MIV stress prediction model and parameter setting

In this paper, we develop a two-layer 3D IC model in COMSOL based on the MIV structure. By simulating various temperature distributions, we obtain the thermal stress data for the chip. We then establish a 3D IC stress prediction model by coupling the electrical, thermal, and mechanical fields. From Fig. 16, we can clearly see a two-layer chip structure model connected by MIVs. In COMSOL, we carry out multi-physics simulation, and the boundary conditions of the chip model can be flexibly adjusted as needed. The whole chip model is set to be placed at room temperature. The bottom surface is set as a fixed constraint condition. The entire chip is divided into 40 blocks of the same size, representing 40 different cores. A different temperature is applied to the bottom of each cores to obtain the corresponding amount of stress.

3.2. Artificial neural network stress prediction model

The stress in 3D ICs is related to temperature, as shown by finite element simulation. [28]. Therefore, in order to predict the thermal mechanical stress caused by temperature, we propose a BP neural network optimized based on genetic algorithm to improve the operational reliability of 3D integrated circuits, and compare it with the general BP network [29]. The genetic algorithm is adaptive, so that it can search for the best weights and biases more accurately, and can quickly adapt to the changes in data distribution, and has good robustness, which can prevent the accuracy of the model from being unacceptable when the neural network input data is abnormal.

The common BP neural network is a multi-layer feed forward neural network trained according to the error back propagation algorithm [30,31]. The most commonly used infrastructure consists of an input layer-hidden layer-output layer, as shown in Fig. 17 and Table 3.

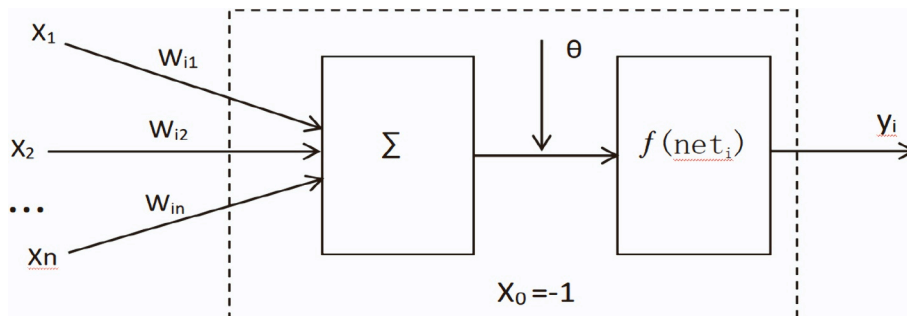
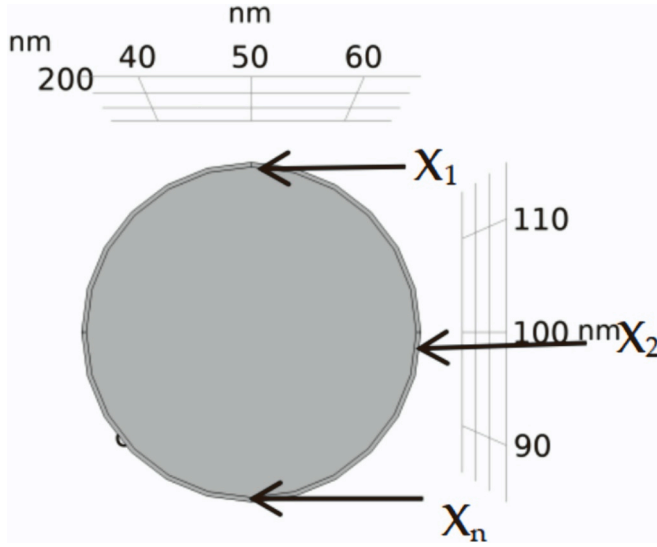


Fig. 17. Neural network diagram.

Table 3

Neural network stress model algorithm flow.

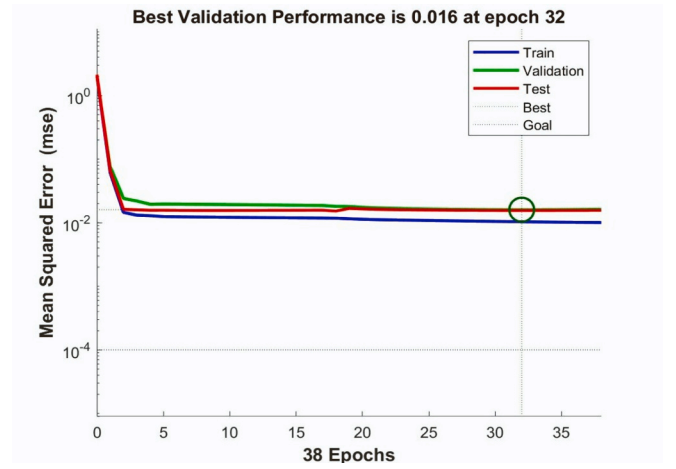
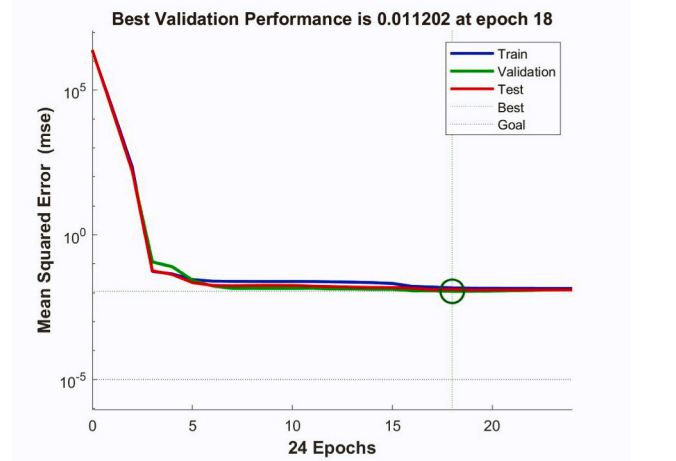
Algorithm stress data prediction algorithm based on BP and GABP neural network
1. Three-dimensional integrated circuit stress model calculation.
2. Obtaining temperature and stress data.
3. The data is divided into test set and training set.
4. Taking the temperature information as input, the stress information is calculated by BP stress model and GABP stress model.
5. Take the stress and temperature information as input, calculate the output maximum stress.
6. The training results of the two neural networks are compared with the actual results of COMSOL simulation.
7. Continuously improve the neural network training parameters until the most accurate results are achieved.
8. Get the results, end the training.

**Fig. 18.** Temperature and stress input data.

In this paper, the global optimization ability of genetic algorithm is used to optimize the BP neural network. The key is to optimize the parameters of BP neural network and find the best W_i and θ weight between input layer and hidden layer and between hidden layer and output layer.

Using the BP neural network improved by genetic algorithm stress prediction model, the temperature near MIV is set as the input of the network, and the corresponding maximum thermal stress is set as the output of the network. The whole process of stress training model is as shown in Table 3. Selecting the maximum stress around the MIV is the most important factor affecting reliability to save the computational calculation of the artificial neural network model. So in Fig. 17, X_1 , X_2 , ... X_n is the spatially distributed temperature value near a MIV, and y_i denotes the maximum stress caused by the spatially distributed temperature near MIV. For the MIV, we will have a spatially distributed temperature vector $X_n = \{X_1, X_2, \dots, X_n\}$, and a maximum stress y_i . We call such data (X_n, y_i) a sample. Since the regular-shaped MIV structure in COMSOL is connected to two layers of chips, we can obtain the sample X_n from the same position of each MIV (relative to the center of the MIV) to obtain enough data for the prediction calculation in the artificial neural network. Fig. 18 shows the location of the extracted temperature and stress values in the MIV.

In order to obtain temperature and thermal stress data, a 3D IC model including 15×16 MIVs was established in COMSOL version 5.6. We applied different temperature excitations to the bottom of 40 cores of the 3D IC stress prediction model for multiple calculations. We discuss that the filling material of the MIV structure has stronger thermal conductivity than the chip substrate material silicon, when the high temperature flows through the silicon chip and MIV, the temperature can quickly pass through the metal materials such as copper in the MIV, resulting in a sudden change in the temperature at the edge of the MIV structure,

**(a)****(b)****Fig. 19.** Analysis of error and iteration times: (a) error before GA optimization; (b) error after GA optimization.

that is, there may be a large temperature gradient or a large thermal stress area. Therefore, we select the area near the MIV of the chip to obtain the temperature and stress as the data set for neural network training. There are 240 MIVs on the chip, through multiple simulations, we obtained a total of about 1000 samples. The samples were divided into two parts, 970 samples were used to train and establish the neural network model, and another 30 samples were used to verify the accuracy of the neural network model.

As shown in Fig. 19, we verify the prediction MSE of the stress value in the BP neural network and the genetic algorithm improved BP neural network (GABP). It can be clearly obtained that before the genetic

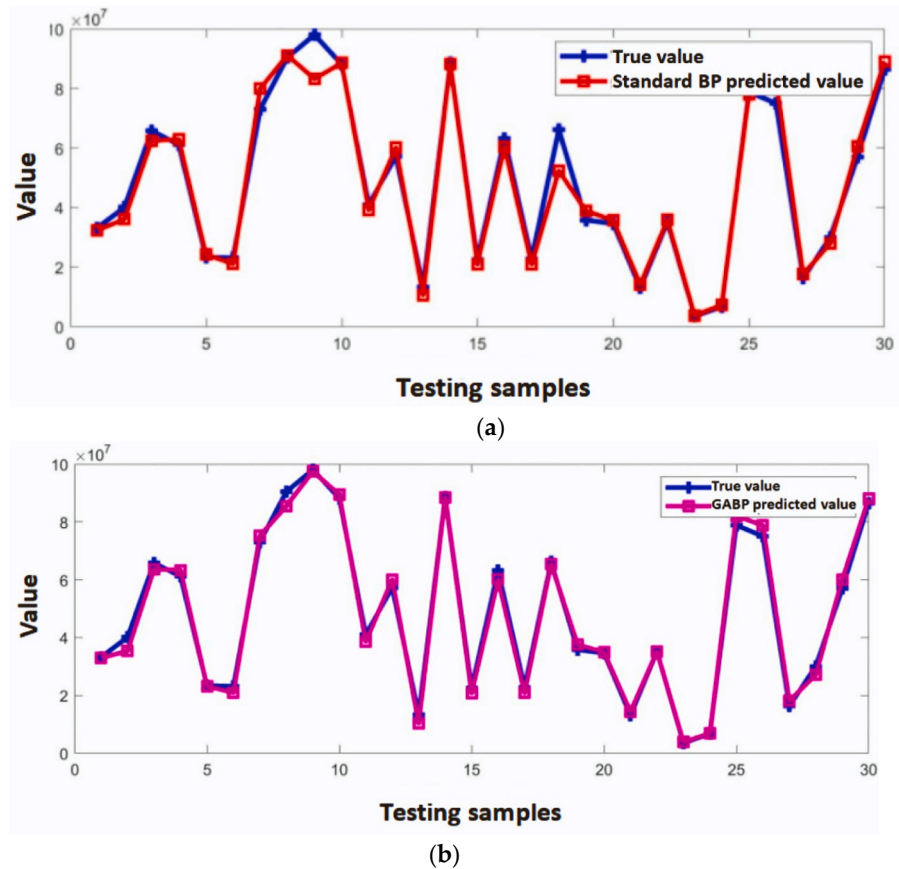


Fig. 20. The verification results of two prediction networks: (a) Standard BP network predicted value and true value; (b) GABP network predicted value and true value.

Table 4

Comparison of the performance of two neural networks.

	BP neural network	GABP neural network
The number of iterations	38Epochs	24Epochs
Mean Squared Error	0.016	0.011
Average error	0.17	0.09

algorithm optimizes the BP network, the number of iterations is about 38 times. After optimization, it is cut off after 24 iterations, indicating that the convergence speed after optimization is significantly improved. It can be seen from the diagram that the mean square error after optimization is also significantly lower than that before optimization.

The results shown in Fig. 20 show the comparison of the predicted stress value of the general BP neural network and the predicted stress value of the BP neural network improved by the genetic algorithm with the true value. We can see that the predicted values of the two neural networks are roughly matched with the true values, but the accuracy of the stress prediction of the BP neural network optimized by the genetic algorithm is still higher than that of the general BP neural network, which is reflected in the samples 9, 18 and 19. In GA-BP, the genetic algorithm is adaptive, and the genetic algorithm adapts to specific problems and datasets through selection, crossover, and mutation operations, which is adaptive, so that the best weights and biases can be searched out more accurately, and the weights and thresholds of the BP neural network can be optimized to obtain more accurate predictions.

Table 4 presents a comparison between the number of iterations, Mean Squared Error (MSE), and average error of the BP and GABP neural networks to achieve optimal training performance. The GABP neural networks demonstrate superior performance characterized by

smaller MSE, fewer iterations required to attain optimal training, and reduced average errors. We attribute this enhanced performance to the GABP neural networks' capabilities.

4. Conclusion

MIV technology offers significant benefits such as heterogeneous integration and reduced signal delay. However, its stacked chip form and interlayer material present challenges. This paper systematically studies the thermal characteristics of MIV using COMSOL finite element simulation, analyzing crosstalk and stress issues, and optimizing parameters like MIV height and radius. Additionally, a stress prediction model based on a neural network and coupled electro-thermal-mechanical fields is proposed. The results indicate that MIV structures with smaller radii and heights exhibit higher thermal reliability. The neural network-based stress prediction model demonstrates high accuracy, providing robust technical support for 3D IC applications.

CRediT authorship contribution statement

Conceptualization: Zhiyuan Zhu; Data curation: Sixiang Zhang; Formal analysis: Sixiang Zhang, Qiuping Yang; Investigation: Sixiang Zhang; Supervision: Zhiyuan Zhu; Writing - original draft: Sixiang Zhang, Qiuping Yang, Zhiyuan Zhu.

Declaration of competing interest

The authors declare that there is no conflict of interest.

Data availability

Data will be made available on request.

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